

# Exact cofactor capacity and negative-shape channel returns for the Erdős–Straus equation

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## Abstract

For  $h = 4j - 1 > t > 0$  we classify every divisor  $W \mid j^2$  in the class  $t \pmod{h}$ . Besides the canonical divisor and an unbounded companion, all words arise from an explicit divisor box on  $\ell \leq (t - 2)/4$ . They satisfy the sharp bound  $h \leq t^2 - 3t + 1$ , which is realized by original exterior states at infinitely many hard primes. This yields exact original middle returns and proves automatic availability on three alpha and nine beta shape lines. For every fixed alpha coefficient  $t \geq 4$ , a separate reduced prime progression has actual exterior states but empty complete same-grade middle boxes. No universal ES occupancy is asserted.

## 1 The precise continuation

This tranche completes a fixed-target divisor problem in the original Type-II cofactor box. It does not assert occupancy at every prime. Starting from an actual exterior state with a negative-four-integer shape, it gives the entire same-grade middle return, including every noncanonical divisor and its inverse. The noncanonical repair has a sharp finite bound depending only on the shape coefficient. That bound is realized by original exterior states at infinitely many hard primes. Conversely, for every fixed shape coefficient  $t \geq 4$ , an explicit reduced prime progression gives original exterior states whose entire same-grade middle box is empty. Their known exterior solutions remain present.

The Type-I/II coordinate identities and the square-divisor cofactor criterion are antecedents [1, §2, Propositions 2.2 and 2.6]. The preceding source [4, negative-four-square transfer] supplied  $t = c^2$  when  $c \mid (h + 1)/4$ . Here the arithmetic restriction is neither omitted nor made into an unspecified admissibility hypothesis: it is solved exactly as a divisor problem. No historical-priority determination is claimed.

## 2 Original coordinates and a general negative shape

Let  $p \equiv 1 \pmod{4}$  be prime. An original exterior state consists of

$$p/4 < a < p/2, \quad R = 4a - p, \quad u \mid a^2, \quad R \mid 4u + 1.$$

Retain  $a = \prod_l l^{e_l}$ , the full box  $-e_l \leq \beta_l \leq e_l$ , and  $u = \prod_l l^{e_l + \beta_l}$ . Set

$$g = \gcd(a, u), \quad (h, r, s) = (g^2/u, u/g, a/g), \quad \kappa = (pr + s)/R.$$

Then  $a = hrs$ ,  $u = hr^2$ ,  $\gcd(r, s) = 1$ , and the ordered denominators are

$$(x, y, z) = (a, hsk, phr\kappa), \quad \frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \tag{1}$$

Multiplication proves the identity, and its ordered inverse is  $u = a^2/(Ry - pa)$ . All normalization factors are positive integers. Let

$$v = hs^2 = a^2/u, \quad D = (4u + 1)/R = (r + \kappa)/s, \quad \alpha = v - R, \quad \beta = v - D.$$

The symbol  $\beta$  here is not an exponent  $\beta_l$ . The original identities are

$$\begin{aligned} RD &= 4hr^2 + 1, & pD + 1 &= 4hr\kappa, & r + \kappa &= sD, \\ h &\mid \alpha\beta - 1, & p &\equiv \alpha \pmod{h}, & p\beta &\equiv 1 \pmod{h}. \end{aligned} \quad (2)$$

For every original divisor  $d \mid a^2$ , reciprocity gives  $(d/R) = (d/p)$ . For an odd prime  $l \mid a$ , use  $R \equiv -p \pmod{l}$  and  $R \equiv 3 \pmod{4}$ ; for 2 use the supplementary law. The exterior target  $u \equiv -1/4 \pmod{R}$  therefore gives  $(h/p) = (u/p) = -1$ . These are the classical laws, used in their original factor coordinates [2, §6.4].

Suppose one ordered shape coordinate is  $-4t$ ,  $t \geq 1$ . Since  $R, D \equiv 3 \pmod{4}$ ,

$$h \equiv 3 \pmod{4}, \quad s \text{ odd}, \quad \gcd(h, t) = 1, \quad (t/h) = 1. \quad (3)$$

Indeed  $hs^2$  is 3 modulo four. Equation (2) makes  $h$  coprime to  $4t$ . For  $\alpha$ , reciprocity gives  $(h/p) = (p/h) = (-4t/h) = -(t/h)$ ; the inverse congruence gives the same result for  $\beta$ . Also  $p > 2h$  because  $a \geq h$ .

Put  $j = (h + 1)/4$  and define the full original target set

$$\boxed{\mathcal{A}_t(j) = \{W > 0 : W \mid j^2, W \equiv t \pmod{4j - 1}\}}. \quad (4)$$

There is no primality condition on  $h = 4j - 1$  in (4).

**Theorem 2.1** (Complete same-grade channel return). *For every original exterior state as above and every  $W \in \mathcal{A}_t(j)$ , set*

$$U = \begin{cases} W, & \alpha = -4t, \\ j^2/W, & \beta = -4t, \end{cases} \quad Q = h, \quad R_M = (p + 4U)/h, \quad a_M = (p + R_M)/4. \quad (5)$$

*These are exactly the original cofactor- $h$  middle choices, with the indicated ordered shape branch retained. In particular  $U \mid j^2$ ,  $0 < R_M < p$ , and*

$$g_M = \gcd(j, U), \quad h_M = g_M^2/U, \quad r_M = U/g_M, \quad \lambda_M = j/g_M, \quad s_M = R_M\lambda_M - r_M \quad (6)$$

*returns the original middle state and its ordered denominators*

$$(a_M, ph_Ms_M\lambda_M, ph_Mr_M\lambda_M).$$

*Proof.* In the  $\alpha$  case  $p \equiv -4t \pmod{h}$ , so the cofactor congruence is exactly  $W \equiv t$ . In the  $\beta$  case  $p \equiv -(4t)^{-1} \pmod{h}$  and  $j^2 \equiv 16^{-1} \pmod{h}$ , so  $U = j^2/W$  satisfies  $p + 4U \equiv 0$  precisely when  $W \equiv t$ . Every divisor involved is a unit modulo  $h$ . Since  $U \leq j^2$ ,  $h \geq 3$  and  $p > 2h$ ,

$$4U \leq (h + 1)^2/4 < (h - 1)p.$$

Thus  $R_M < p$ , and its positive integer value is 3 modulo four. Prime valuations in  $U \mid j^2$  give  $h_Mr_M\lambda_M = j$ ,  $h_Mr_M^2 = U$ , and  $\gcd(r_M, \lambda_M) = 1$ . Directly

$$a_M = R_Mj - U = h_Mr_Ms_M > 0.$$

A common prime of  $r_M$  and  $R_M$  would divide  $p$  by  $hR_M = p + 4U$ , whereas  $r_M \leq j < p$ . Hence  $\gcd(r_M, s_M) = 1$ . These identities prove the middle normalization, the first-half range, and the reciprocal identity. Conversely a cofactor- $h$  middle word has  $U \mid j^2$  and the same gate; taking  $W = U$  or  $j^2/U$  recovers (4). Neither  $U < a_M$  nor  $r_M < s_M$  is imposed: the ordered orientation is kept, or complemented explicitly afterward.  $\square$

### 3 An exact disjoint classification of every available divisor

The next theorem is pure integer arithmetic. In particular,  $t$  need not be a square and  $h$  need not be prime.

**Theorem 3.1** (Fixed-target capacity completion). *Let  $t, j \geq 1$  and  $h = 4j - 1 > t$ . The set  $\mathcal{A}_t(j)$  is the disjoint union of:*

1. *the canonical word  $W = t$ , present exactly when  $t \mid j^2$ ;*
2. *the companion word  $W = 4tj$ , present exactly when  $4t \mid j$ ;*
3. *the finite words described by*

$$1 \leq \ell \leq \left\lfloor \frac{t-2}{4} \right\rfloor, \quad w \mid \ell^2, \quad n = \frac{t-w}{4\ell+1} \in \mathbb{Z}_{>0}, \quad (7)$$

$$V = \ell^2/w, \quad j = \ell + 4nV, \quad h = 4j - 1 > t, \quad W = j^2/V = t + nh. \quad (8)$$

*In the last item the displayed  $j$  must equal the input  $j$ . Each word occurs once. The inverse, for  $W \neq t$ , is*

$$n = (W - t)/h, \quad V = j^2/W, \quad \ell = j - 4nV. \quad (9)$$

*For  $\ell = 0$  it is the companion. Otherwise  $w = \ell^2/V$  is the unique label in (7).*

*Proof.* If  $W \neq t$ , then  $W = t + nh$  for some  $n \geq 1$ , because  $0 < t < h$ . Let  $V = j^2/W$ . Reduction of  $16WV = 16j^2 = (h+1)^2$  modulo  $h$  gives

$$16tV = 1 + kh$$

for an integer  $k \geq 1$ . Modulo four,  $k \equiv 1$ , so  $k = 4\ell + 1$  with  $\ell \geq 0$ . Comparing the two exact products gives

$$h + 2 = k + 16nV, \quad j = \ell + 4nV, \quad t = (4\ell + 1)n + \frac{\ell^2}{V}. \quad (10)$$

If  $\ell = 0$ , then  $n = t$ ,  $j = 4tV$ , and  $W = 4tj$ . This is equivalent to  $4t \mid j$  and coexists with the canonical divisor. If  $\ell > 0$ , (10) proves  $V \mid \ell^2$ , and  $w = \ell^2/V \geq 1$ . Therefore  $t \geq 4\ell + 2$ , giving (7). All labels are recovered by (9). Conversely (7)–(8) give

$$j^2/V = w + 8\ell n + 16n^2V = t + n(4j - 1),$$

an integer divisor of  $j^2$  in the required class. The three cases are disjoint by  $W = t$ ,  $\ell = 0$ , or  $\ell > 0$ .  $\square$

The small-grade domain  $h \leq t$  is separately finite:  $j \leq \lfloor (t+1)/4 \rfloor$ . Literal enumeration there is retained; the theorem is not asserted on that domain. A reduced residue  $t \bmod h$  may be used to test a word, but it must not replace the original shape coefficient in a bound.

**Theorem 3.2** (Sharp finite-repair bound). *Every finite word in Theorem 3.1 satisfies*

$$\boxed{h \leq t^2 - 3t + 1.} \quad (11)$$

*Equality holds exactly when  $w = n = 1$  and  $t = 4\ell + 2$ . In particular, if the canonical divisor is absent but  $\mathcal{A}_t(j)$  is nonempty, (11) holds. For  $h > \max(t, t^2 - 3t + 1)$  the exact count is*

$$\boxed{|\mathcal{A}_t(j)| = \mathbf{1}_{t \mid j^2} + \mathbf{1}_{4t \mid j}.} \quad (12)$$

*Proof.* Use  $n = (t - w)/(4\ell + 1)$  to write

$$h + 1 = 4\ell + \frac{16\ell^2(t - w)}{(4\ell + 1)w} \leq 4\ell + \frac{16\ell^2(t - 1)}{4\ell + 1}.$$

The last expression is strictly increasing in real  $\ell \geq 0$ . Since  $\ell \leq (t - 2)/4$ , it is at most  $(t - 2) + (t - 2)^2 = t^2 - 3t + 2$ . Equality requires both  $w = 1$  and  $\ell = (t - 2)/4$ , and then  $n = 1$ . The companion cannot repair a missing canonical word because  $4t \mid j$  implies  $t \mid j^2$ . The classification proves (12).  $\square$

For every  $\ell \geq 1$  the arithmetic equality family is

$$t = 4\ell + 2, \quad j = \ell(4\ell + 1), \quad h = 4j - 1, \quad W = (4\ell + 1)^2, \quad V = \ell^2. \quad (13)$$

Here  $t \nmid j^2$ : the odd number  $2\ell + 1 > 1$  is coprime to both factors of  $j$ . Thus (13) is a genuine repair, not a second name for the canonical word. Not every finite word is a repair: at

$$t = 289, \quad j = 1224, \quad h = 4895, \quad \mathcal{A}_t(j) = \{289, 5184\},$$

the canonical word and a finite word coexist. The latter has  $(n, V, \ell, w) = (1, 289, 68, 16)$ . Its coefficient is two, even though its occupied residue is a single class.

The complete finite search uses at most

$$\sum_{\ell \leq (t-2)/4} \tau_{\text{div}}(\ell^2) \leq L(1 + \log L)^2, \quad L = \lfloor (t - 2)/4 \rfloor \geq 1. \quad (14)$$

Indeed  $\tau_{\text{div}}(\ell^2) \leq d_3(\ell)$  primewise, and  $\sum_{abc \leq L} 1 \leq L(\sum_{a \leq L} 1/a)^2$ . This counts candidate labels under unit-cost integer arithmetic; it is not a bit-complexity estimate that includes integer factorization. It is a finite classification in the original integers, not an assertion that a successful word always exists.

There is also a strict smaller-parameter return on each finite word. Put  $g_0 = \gcd(\ell, V)$  and

$$H_0 = g_0^2/V, \quad r_0 = V/g_0, \quad \lambda_0 = \ell/g_0.$$

Then  $\ell = H_0 r_0 \lambda_0$ ,  $V = H_0 r_0^2$ , and  $\gcd(r_0, \lambda_0) = 1$ . The middle normalization in (6) is

$$\begin{array}{c|ccc} & h_M & r_M & \lambda_M \\ \hline \alpha & H_0 & \lambda_0 + 4nr_0 & r_0 \\ \beta & H_0 & r_0 & \lambda_0 + 4nr_0. \end{array} \quad (15)$$

In particular  $h_M \leq \ell \leq (t - 2)/4$ . This decreases a parameter of the finite capacity reduction; it is not a well-founded proof of initial ES occupancy.

## 4 Entire shape lines with forced original availability

**Theorem 4.1** (Automatic negative-shape returns). *Let  $p \equiv 1 \pmod{24}$  be prime and start from an original exterior state. The canonical word is always available on the three lines*

$$\alpha = -4t, \quad t \in \{1, 2, 3\},$$

*and on the nine lines*

$$\boxed{\beta = -4t, \quad t \mid 36.} \quad (16)$$

*Every such line has the original middle return (5)–(6), with no bound on the other shape coordinate. The coefficient  $t$  need not be a square.*

*Proof.* For  $\alpha$ , (3) gives  $(2/h) = 1$  at  $t = 2$ ; together with  $h \equiv 3 \pmod{4}$ , this forces  $h \equiv 7 \pmod{8}$ , hence  $2 \mid j$ . At  $t = 3$ , the supplementary quadratic law gives  $h \equiv 11 \pmod{12}$  and  $3 \mid j$ . The case  $t = 1$  is literal.

For  $\beta$ , use the stronger actual equations, not just their Jacobi projection. Here  $D = hs^2 + 4t$  and  $r + \kappa = sD$  is odd. Thus  $r\kappa$  is even. Since  $pD + 1 = 4hr\kappa$  and  $p \equiv 1 \pmod{8}$ , necessarily  $D \equiv 7 \pmod{8}$ , whence

$$h \equiv 7 - 4t \pmod{8}.$$

If  $2 \mid t$ , this gives  $2 \mid j$ . If  $3 \mid t$ , all equalities in the rest of this paragraph are modulo three. The case  $h = 0$  makes  $D = 0$  and contradicts  $pD + 1 = 4hr\kappa$ . If  $h = 1$  and  $s = 0$ , the same equation would give  $1 = -r^2$ , impossible. If  $h = 1$  and  $s = \pm 1$ , then  $D = 1$ ,  $\kappa = s - r$ , and  $2 = r(s - r)$ ; the right side has only values zero and one. Thus  $h = 2 \pmod{3}$ , so  $3 \mid j$ . Every prime exponent of  $t \mid 36$  is at most two. Therefore these two forced divisibilities imply  $t \mid j^2$ , and Theorem 2.1 applies.  $\square$

For the canonical  $W = t$ , the squarefree part of  $h_M$  is exactly that of  $t$ . This follows from  $h_M = g_M^2/U$  and, in the inverse branch,  $U = j^2/t$ . The preceding square-target theorem is the special case in which this part is one. We do not claim that (16) lists every possible automatic  $\beta$  coefficient. Actual complete failures at  $t = 5, 8, 27$  are given below.

## 5 The quadratic bound is sharp on actual hard-prime sources

**Theorem 5.1** (Prime realization of the finite-repair boundary). *For every  $t = 6 + 420z$ ,  $z \geq 0$ , infinitely many primes  $p \equiv 1 \pmod{840}$  have an original exterior state with  $\alpha = -4t$ ,*

$$h = t^2 - 3t + 1, \quad R = t^2 + t + 1, \quad s = 1,$$

*whose canonical cofactor word is unavailable but whose finite word in (13) returns an original middle state. Thus the bound (11) is attained within the original prime arithmetic, for unbounded  $t$ .*

*Proof.* Put  $\ell = (t - 2)/4$ ,  $j = \ell(4\ell + 1)$ , so  $h = 4j - 1$  and the word is  $W = (t - 1)^2$ . The two integers  $h, R$  are coprime, since their difference is  $4t$  and  $h \equiv R \equiv 1 \pmod{t}$ . For the displayed progression of  $t$ ,  $\gcd(hR, 210) = 1$ . Also  $t^3 \equiv 1 \pmod{R}$ . Consequently

$$r_0 = (t + 2R)/4 \in \mathbb{Z}, \quad 16tr_0^2 \equiv 1 \pmod{R}.$$

Since  $h \equiv -4t \pmod{R}$ , the latter is exactly  $R \mid 4hr_0^2 + 1$ . Choose  $r$  by the explicit compatible CRT conditions

$$r \equiv r_0 \pmod{R}, \quad r \equiv \frac{R+1}{4}h^{-1} \pmod{210}.$$

Then  $p = 4hr - R$  varies in a progression of modulus  $840hR$ , is 1 modulo 840, and is coprime to the entire modulus: modulo  $h$  it is  $-R$  and modulo  $R$  it is  $4hr_0$ . Dirichlet's theorem supplies infinitely many prime members [3, Theorem 18.1]. For all sufficiently large positive members,  $a = hr$ ,  $u = hr^2$  is an original first-half exterior state;  $R < 2hr$  verifies the only upper range issue. Its shape is  $h - R = -4t$ . Formula (13) proves canonical unavailability and supplies the sharp finite repair. The target is reconstructed by Theorem 2.1. Prime values are supplied by a reduced *linear* progression; no assertion about primes represented by a higher-degree polynomial has been inserted.  $\square$

A small explicit original member, in another hard class, is

$$\boxed{p = 825241, \quad (a, R, u, h, r, s, \kappa) = (206321, 43, 2240439739, 19, 10859, 1, 208402140).} \quad (17)$$

Its coefficient is  $t = 6$ ,  $j = 5$ , and the entire capacity box is  $\{1, 5, 25\}$ . Only  $25 \equiv 6 \pmod{19}$  succeeds. The original middle return is

$$(a_M, R_M, U, h_M, r_M, s_M, \lambda_M) = (217170, 43439, 25, 1, 5, 43434, 1), \quad (18)$$

with ordered denominators  $(217170, 825241 \cdot 43434, 825241 \cdot 5)$ . This is a nonsquare coefficient and a genuinely noncanonical return.

## 6 Every other alpha coefficient has infinite original same-grade obstructions

**Lemma 6.1** (Exact limitation of the character-only budget). *For  $t \geq 1$ , let the product run over every prime, including 2:*

$$K(t) = \prod_{l^e \parallel t} l^{\lceil e/2 \rceil}, \quad M = 4K(t).$$

*For odd  $A$  coprime to  $t$ , write  $\chi_t(A) = (t/A)$  for the quadratic Kronecker–Jacobi character determined by the squarefree part of  $t$ ; its conductor divides  $M$ . Among units  $A \pmod{M}$  with  $A \equiv 3 \pmod{4}$  and  $\chi_t(A) = 1$ , the implication  $A \equiv -1 \pmod{M}$  holds for all such  $A$  exactly when  $t = 1, 2, 3$ .*

*Proof.* The Jacobi character is well-defined modulo  $M$  by reciprocity, including the factor at two. It takes value one at  $-1$ . If  $t$  is a square, it is trivial, and the indicated set has  $\varphi(M)/2$  elements. If  $t$  is nonsquare, its restriction to the units 1 modulo four is nontrivial: at an odd prime appearing to odd exponent choose a nonsquare by CRT, with the other coordinates one; if only the factor two has odd exponent, choose the coordinate 5 modulo eight. Thus it is independent of the mod-four sign character, and the indicated set has  $\varphi(M)/4$  elements. A multiple of four with totient two is  $M = 4$ ; those with totient four are 8, 12. The square/non-square alternatives give exactly  $t = 1, 2, 3$ . In each singleton case the member is  $-1$ .  $\square$

**Theorem 6.2** (An actual obstruction for every fixed  $t \geq 4$ ). *For every integer  $t \geq 4$ , there are infinitely many hard primes possessing an original exterior state with  $\alpha = -4t$  but no middle state at cofactor equal to that state’s original grade  $h$ , even after the entire divisor box  $\text{Div}(j^2)$  is tested. These primes already have the constructed exterior solution. This is not an Erdős–Straus counterexample.*

*Proof.* Choose a unit class  $A \pmod{4K(t)}$  from the preceding lemma other than  $-1$ . Then  $A \equiv 3 \pmod{4}$  and  $(t/A) = 1$ . Lift it by CRT to a reduced class modulo  $\text{lcm}(4K(t), 840)$  such that, for  $l = 3, 5, 7$ , both  $R$  and  $R - 4t$  are units modulo  $l$ . If  $l \mid K(t)$  this is already true, because  $l \mid t$  and  $A$  is a unit. Otherwise one need only avoid  $0, 4t \pmod{l}$ , which leaves a choice even for  $l = 3$ . Choose an odd lift modulo eight as well. Dirichlet’s theorem gives a prime  $R$  in this class as large as required. Set  $h = R - 4t$  and choose it so that

$$h > \max(4t, t^2 - 3t + 1).$$

Then  $h \equiv 3 \pmod{4}$ ,  $\gcd(h, 210) = \gcd(h, R) = 1$ , and  $h \not\equiv -1 \pmod{4K(t)}$ . Thus  $t \nmid j^2$  for  $j = (h + 1)/4$ . Since  $\chi_t(R) = (t/R) = 1$ , choose  $q$  with  $q^2 \equiv t \pmod{R}$  and set  $r_0 \equiv (4q)^{-1} \pmod{R}$ . Then

$$16tr_0^2 \equiv 1 \pmod{R}.$$

Solve

$$r \equiv r_0 \pmod{R}, \quad r \equiv \frac{R+1}{4} h^{-1} \pmod{210}.$$

Again  $p = 4hr - R$  is a reduced progression modulo  $840hR$ , with  $p \equiv 1 \pmod{840}$ . Dirichlet gives infinitely many prime members. Take  $a = hr$ ,  $u = hr^2$ ,  $s = 1$ . Then  $R \mid 4u + 1$ ,  $\gcd(p, a) = 1$ ,

and  $p > 2a$  follows already from  $h > 4t$  and  $r \geq 1$ . The original shape is  $\alpha = h - R = -4t$ . Theorem 3.2 now gives  $\mathcal{A}_t(j) = \emptyset$ : neither the canonical word nor its companion is available and the finite-exception range has been passed. Theorem 2.1 identifies this with failure of the complete cofactor- $h$  middle source.  $\square$

For  $t = 4$  one explicit construction is

$$R = 59, \quad h = 43, \quad j = 11, \quad r = 8415 + 12390n, \quad \boxed{p = 1447321 + 2131080n.} \quad (19)$$

The progression is reduced and 1 modulo 840. At  $n = 3$ ,  $p = 7840561$  is prime, with

$$(a, R, u, h, r, s, \kappa) = (1960155, 59, 89353665675, 43, 45585, 1, 6057830054).$$

The three capacity divisors are 1, 11, 121, of residues 1, 11, 35 modulo 43, not four. Every prime in (19) nevertheless has its displayed exterior solution.

The beta automatic theorem has genuine limits too. The following are original hard-prime sources, and complete enumeration of their small  $j^2$  boxes proves failure at their old grade:

| $p$     | $a$    | $u$       | $h$ | $r$  | $s$ | $\kappa$ | $t \ (\beta = -4t)$ |
|---------|--------|-----------|-----|------|-----|----------|---------------------|
| 346201  | 158815 | 43864703  | 23  | 1381 | 5   | 1654     | 8                   |
| 593401  | 188100 | 39710000  | 11  | 1900 | 9   | 7091     | 27                  |
| 2727841 | 929071 | 464321099 | 11  | 6497 | 13  | 17930    | 5.                  |

(20)

The case  $h \leq t$  in the middle row is checked literally, not attributed to Theorem 3.1. No infinite beta obstruction theorem is asserted.

## 7 Original inverse fibres and a necessary prose correction

For fixed  $p, h, t$  on an  $\alpha$  branch, put  $N = (p + 4t)/h$ . Its complete original fibre is

$$s \mid N, \quad s + N/s \equiv 0 \pmod{4}, \quad r = (s + N/s)/4, \quad a = hrs, \quad u = hr^2, \quad \boxed{\gcd(r, s) = 1, \quad p/4 < a < p/2, \quad R = 4a - p \mid 4hr^2 + 1.} \quad (21)$$

The shape equation is  $4rs - s^2 = N$ ; the extra gate in (21) is necessary and not implied by this factorization. For a beta branch, put  $F = (4tp + 1)/h$  and enumerate

$$1 \leq s \leq \lfloor (p - 1)/(2h) \rfloor, \quad D = hs^2 + 4t, \quad \Delta_s = s^2(D^2 - p) - F.$$

Keep the positive squares  $k^2 = \Delta_s$  with  $sD - k$  positive even, and set  $r = (sD - k)/2$ ,  $\kappa = sD - r$ , retaining the original coprimality, range and gate. These are equivalences from  $4hr(sD - r) = pD + 1$  and  $k = \kappa - r > 0$ .

The predecessor's prose alpha-fibre formula omitted the final gate in (21). Its executable inverse-fibre enumeration did test that gate. At  $p = 5209, h = 95, t = 4$ , the displayed divisor equation has  $N = 55$ . The choice  $s = 5, r = 4$  satisfies its factorization, coprimality, shape and first-half conditions, but gives

$$a = 1900, \quad R = 2391, \quad u = 1520, \quad (4u + 1) \bmod R = 1299.$$

It is not an exterior state. This correction changes the textual claimed inverse domain, not the predecessor's forward theorem or its executable fibre filter. The old source is preserved, and **CORRECTION.md** identifies the affected passage and file hash.

For the new maps, retain  $(p, \text{original E state, side, } t, j, W)$  and its canonical/companion/finite label. The M output recovers  $Q = h$  and  $j$ ; (21) and its beta counterpart recover precisely the possible E sources once  $t$  and the side remain. The finite labels are inverted by (9). Across

two ordered branches, alpha word  $W_1$  and beta word  $W_2$  give the same M word exactly when  $W_1W_2 = j^2$ . Multiple original sources above that word are not identified.

On the free abelian group of these labelled sources the coefficient pushforward sums each actual target fibre. Its kernel is generated by same-target differences. No division by a fibre size is used to prove positivity. Reducing a positive even coefficient modulo two can yield a supported zero; it does not remove its original words. The two-word example following (13) is an explicit control.

## 8 Verification scope and the remaining milestone

The verifier exhausts (4) for  $t \leq 120$ ,  $j \leq 5000$ ,  $4j - 1 > t$ , and compares every integer word with Theorem 3.1. It separately verifies all finite profiles and one hundred members of the sharp arithmetic family, the original prime-realizable progressions, and the character-only classification through 120. It also enumerates every original first-half E/M divisor at primes  $p \equiv 1 \pmod{4}$  through 3000, retaining actual factorization and channel records, and checks the returned original states. The fixed larger examples use complete trial-division primality checks. Their empty cofactor boxes are exhausted.

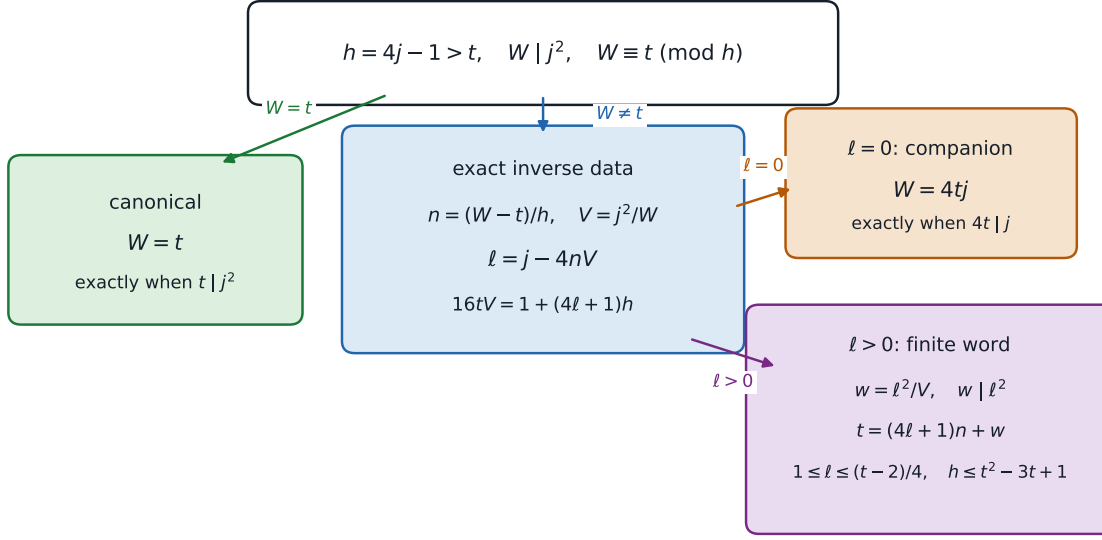
The independent program imports neither the main verifier nor predecessor code. It uses literal divisors of  $j^2$ , coprime primitive  $r, s$  sources, primewise Euler-symbol evaluation, and a complementary-divisor parameter for the finite profiles. Normal and optimized command outcomes, hashes and scopes belong to the execution receipt. Multiple runs are not independent mathematical review.

The completed unbounded statements here concern the *fixed-target capacity problem* and the original shape lines in Theorem 4.1. The obstruction in Theorem 6.2 proves why increasing the old-grade divisor search cannot repair every negative-square alpha source. It does not exclude a different original middle cofactor or an exterior solution, which the construction already supplies. No initial state has been forced at an arbitrary hard prime. Turn 7's universal existence milestone remains outstanding; no Turn 8 witness guarantee, Lean build, new global verification range or priority claim follows.

## References

- [1] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *Journal of the Australian Mathematical Society*, 94(1), 50–105. [arXiv:1107.1010](#). Section 2, Propositions 2.2 and 2.6.
- [2] Hatcher, A. (2022). *Topology of Numbers*. American Mathematical Society. Section 6.4: quadratic reciprocity and supplementary laws.
- [3] Sutherland, A. V. (2017). *18.785 Number Theory I, Lecture 18*. Massachusetts Institute of Technology. Theorem 18.1: Dirichlet's theorem.
- [4] The Clankers. (2026). *The ES–Fabel arithmetic bridge*. Supplied 20 September source, `es_turn7_fabel_bridge_20260920/core.tex`, negative-four-square transfer and its inverse-source subsection. The textual alpha inverse domain is corrected explicitly in the present note; the predecessor executable already retained the missing gate.

### Exact inverse classification of a fixed-target cofactor word



The three outputs are disjoint and the displayed formulas recover every label. The quadratic bound applies only to the finite branch.

Figure 1: The inverse in Theorem 3.1. Starting from the exact word  $W$ , equations (9)–(10) recover  $n, V, \ell$ ; the signs and domains separate the canonical, companion and finite branches. The grade bound belongs only to the finite branch.

### Sharp capacity boundary and its nonsquare channel return

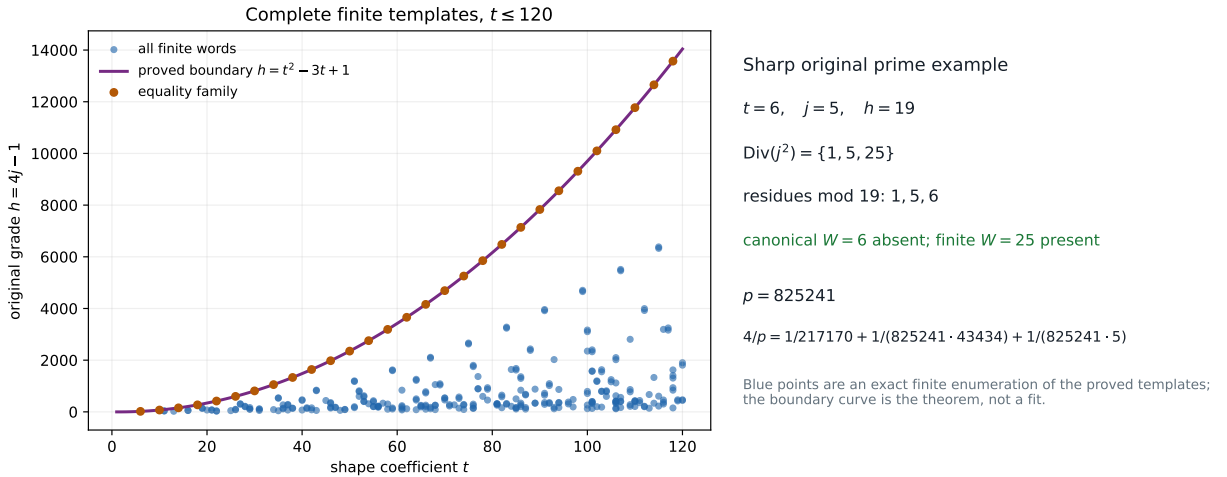


Figure 2: Every finite template with  $t \leq 120$ , plotted against the proved boundary in Theorem 3.2. The right panel retains the exact  $p = 825241$  nonsquare return from equations (17)–(18). The points are a reproducible enumeration by `tools/generate_20260920_illustrations.py`; the curve is the proved bound, not an empirical fit.